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COMPILER DESIGN LAB

SL NO.	TITLE	PAGE NO.	DATE
2a	To identify some keywords, identifiers, integers and real numbers from a simple C program using Lex/Flex.	2-3	19/3/25
2b	To count the number of lines, words, visible characters, total characters using Lex/Flex.	4-5	19/3/25
2ci	To append line number before each lines(empty/non-empty) using Lex/Flex.	6-7	19/3/25
2cii	To append line number before each non-empty lines using Lex/Flex.	7-8	19/3/25
3a	To copy a file by replacing multiple sequences of white spaces with a single white space.	9-10	19/3/25
3b	To copy a file by replacing multiple sequences of white spaces with a single white space and to remove the comments.	11-12	19/3/25

2) a) Using Lex/Flex , write a program to identify some keywords, identifiers, integers and real numbers from a simple C program. Input/output streams may be files.

Lex/Flex Code:

```
%{
#include <stdio.h>
%}

KEYWORD   "int"|"float"|"if"|"else"|"return"|"char"|"void"|"for"|"while"
IDENTIFIER [a-zA-Z_][a-zA-Z0-9_]*
INTEGER   [0-9]+
REAL      [0-9]+"."[0-9]+


%option noyywrap
%%

{KEYWORD}     { printf("Keyword: %s\n", yytext); }
{IDENTIFIER}  { printf("Identifier: %s\n", yytext); }
{INTEGER}     { printf("Integer: %s\n", yytext); }
{REAL}        { printf("Real number: %s\n", yytext); }

[ \t\r\n]     ;        // Ignore whitespace
.             { printf("Symbol: %s\n", yytext); }

%%

int main(int argc, char **argv) {
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile>\n", argv[0]);
        return 1;
    }

    FILE *file = fopen(argv[1], "r");
    if (!file) {
        perror("Failed to open file");
        return 1;
    }

    yyin = file;
    yylex();
```

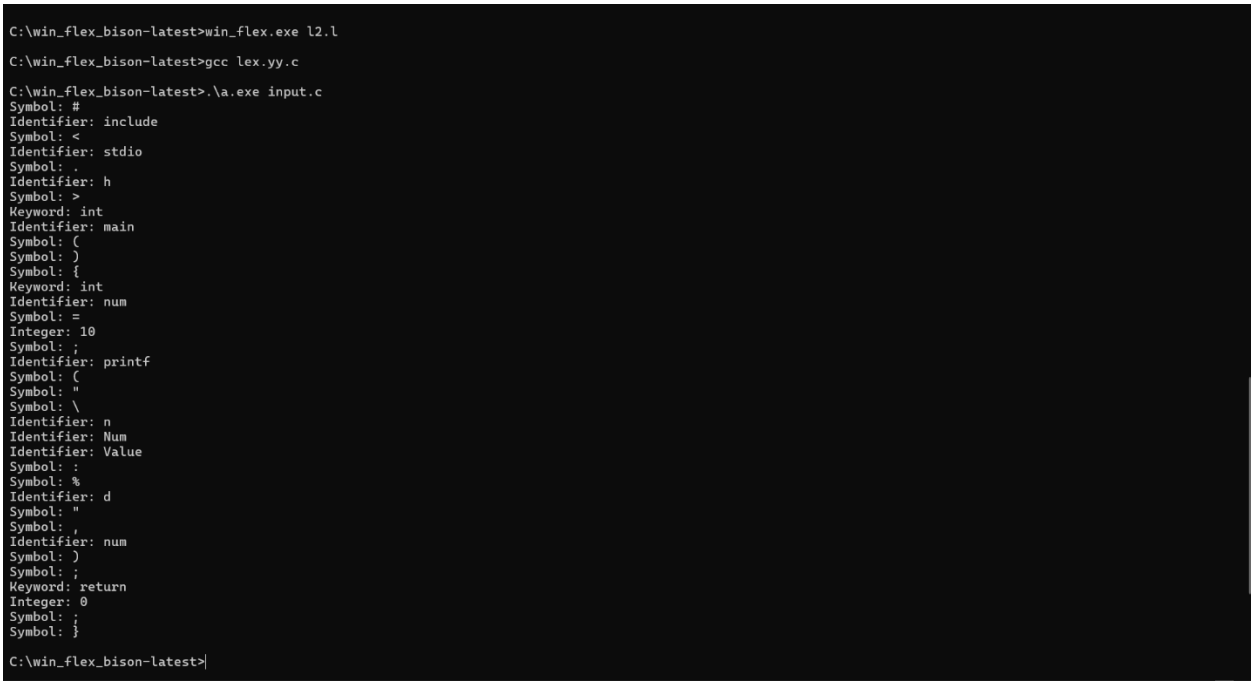
```
    fclose(file);

    return 0;
}
```

Input C program:

```c
#include <stdio.h>
int main() {
    int num = 10;
    printf("\n Num Value: %d", num);
    return 0;
}
```

Output:



b) Using Lex/Flex , write a program to count the number of lines, words, visible characters, total characters.
Input/output streams may be files.

Lex/Flex Code:

```
%{
#include <stdio.h>
int line_count = 0, word_count = 0, visible_count = 0, total_count = 0;
int in_word = 0;  // Flag to track if we are inside a word
%}
%option noyywrap

%%
\n                { line_count++; total_count++; in_word = 0; }
[ \t\r\f]+        { total_count++; in_word = 0; }
[a-zA-Z0-9]+      { visible_count += yyleng; total_count += yyleng; if (!in_word) { word_count++;
in_word = 1; } }
.                 { visible_count++; total_count++; in_word = 0; }
%%

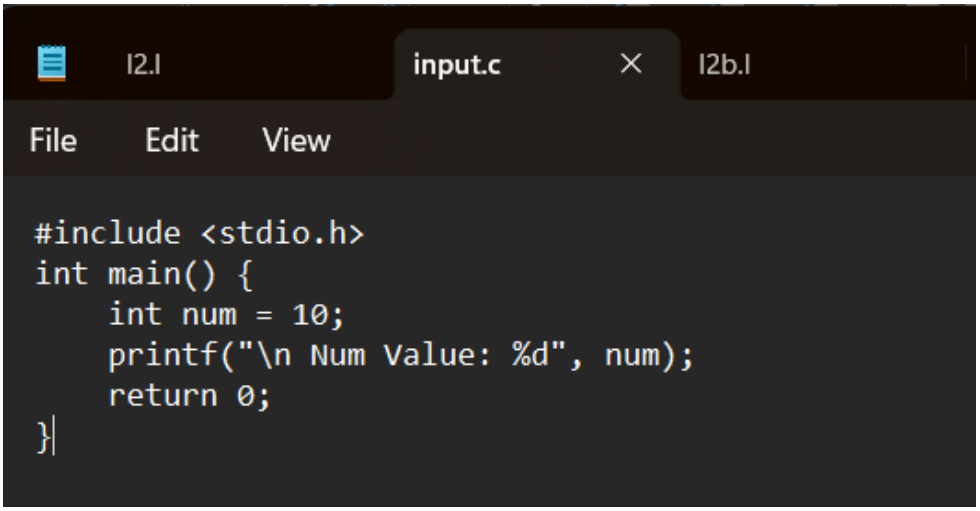
int main(int argc, char **argv) {
    FILE *file;

    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile>\n", argv[0]);
        return 1;
    }

    file = fopen(argv[1], "r");
    if (!file) {
        perror("Failed to open file");
        return 1;
    }
    yyin = file;
    yylex();
    fclose(file);
    printf("\nLines: %d\n", line_count);
    printf("Words: %d\n", word_count);
    printf("Visible Characters: %d\n", visible_count);
    printf("Total Characters: %d\n", total_count);

    return 0;
}
```

Input C program:



Output:

```
C:\win_flex_bison-latest>win_flex l2b.l

C:\win_flex_bison-latest>gcc lex.yy.c

C:\win_flex_bison-latest>.\a.exe input.c

Lines: 5
Words: 16
Visible Characters: 74
Total Characters: 93

C:\win_flex_bison-latest>
```

c) Using Lex/Flex , write a program to append line number before each
(i) lines(empty/non-empty).

Lex/Flex Code:

```
%{
#include <stdio.h>
int line_count = 1;  // Line number counter
%}

%option noyywrap
%%


^(.*)\n?             {
                    printf("%d: %s\n", line_count++, yytext);
                }

.                   { /* Catch-all for any remaining character */ }

%%

int main(int argc, char **argv) {
    FILE *file;
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile.txt>\n", argv[0]);
        return 1;
    }

    file = fopen(argv[1], "r");
    if (!file) {
        perror("Failed to open file");
        return 1;
    }

    yyin = file;
    yylex();
    fclose(file);

    return 0;
}
```

Input Text File:

input.txt

```
Hello, world!

This is a sample text file.
Lex/Flex is fun!

EOF
```

Output:

```
C:\Users\iohhu\Desktop\Compiler design lab\win_flex_bison-latest>win_flex.exe 2c1.l

C:\Users\iohhu\Desktop\Compiler design lab\win_flex_bison-latest>gcc lex.yy.c

C:\Users\iohhu\Desktop\Compiler design lab\win_flex_bison-latest>.\a.exe input.txt
1: Hello, world!

2:

3: This is a sample text file.

4: Lex/Flex is fun!

5:

6: EOF
```

c) Using Lex/Flex , write a program to append line number before each (ii) non-empty lines.

Lex/Flex Code:

```
%{
#include <stdio.h>
int line_count = 1;  // Line number counter
%}
%option noyywrap
%%

^[^\n]+          {
                 printf("%d: %s\n", line_count++, yytext);
               }

^\n              {
                 printf("\n");
               }

.                { /* Catch-all for any remaining character */ }
```

```
%%

int main(int argc, char **argv) {
    FILE *file;

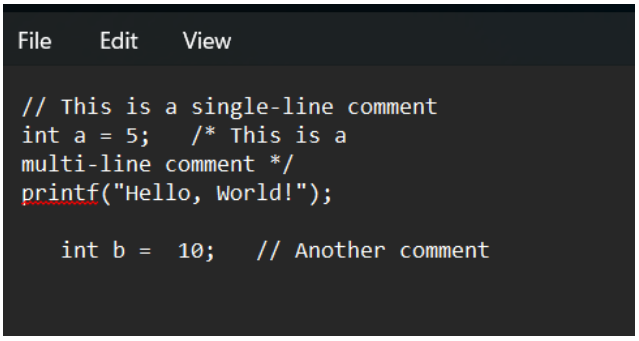
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile.txt>\n", argv[0]);
        return 1;
    }

    file = fopen(argv[1], "r");
    if (!file) {
        perror("Failed to open file");
        return 1;
    }

    yyin = file;
    yylex();
    fclose(file);

    return 0;
}
```

Input Text File:



Output:

```
C:\win_flex_bison-latest>win_flex l2c.l

C:\win_flex_bison-latest>gcc lex.yy.c

C:\win_flex_bison-latest>.\a.exe input.txt
1: // This is a single-line comment

2: int a = 5;   /* This is a

3: multi-line comment */

4: printf("Hello, World!");


5:    int b =  10;   // Another comment
```

3) a) Lex program to copy a file by replacing multiple sequences of white spaces with a single white space.
[blanks/tab => blank, more than one “ \n” => “ \n”].

Lex/Flex Code:

```
%{
#include <stdio.h>
%}

%option noyywrap
%%


[ \t]+              { printf(" "); }

\n+                  { printf("\n"); }


.                   { printf("%s", yytext); }

%%

int main(int argc, char **argv) {
    FILE *infile;

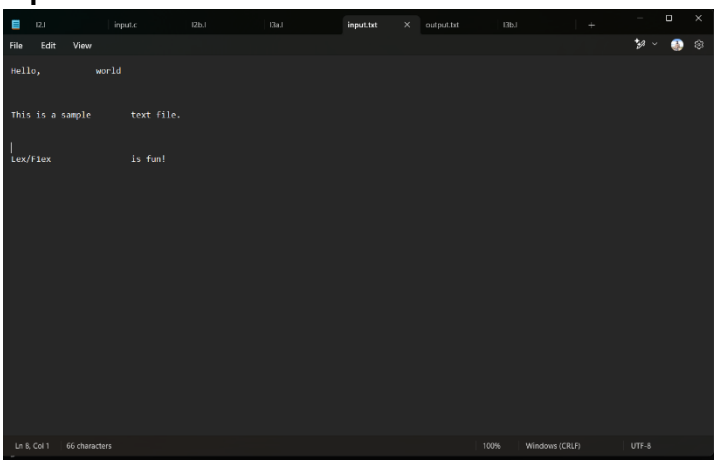
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile.txt>\n", argv[0]);
        return 1;
    }

    infile = fopen(argv[1], "r");
    if (!infile) {
        perror("Failed to open input file");
        return 1;
    }

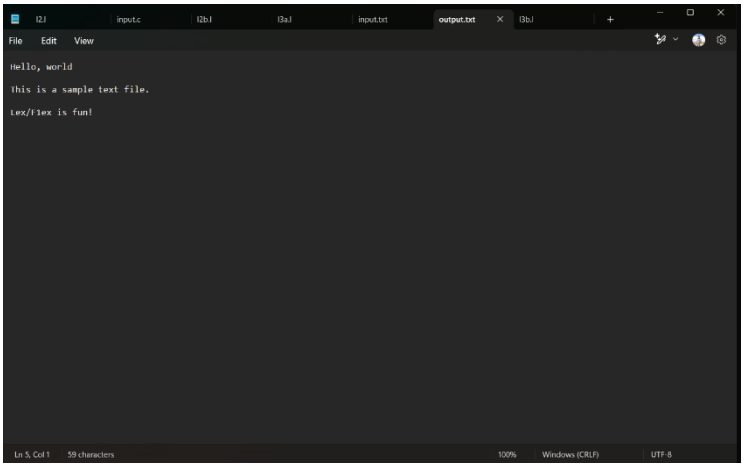
    yyin = infile;
    yylex();
    fclose(infile);

    return 0;
}
```

Input text file:



Output:



b)Lex program to copy a file by replacing multiple sequences of white spaces with a single white space.Also add removal of comments in above program.
[blanks/tab => blank, more than one “ \n” => “ \n”].

Lex/Flex Code:

```
%{
#include <stdio.h>
%}

%option noyywrap

%%

"//".*                      { /* Ignore single-line comments */ }
"/*"([^*]|\*+[^*/])*\*+"/"    { /* Ignore multi-line comments */ }
[\n]+                        { printf("\n"); }
[ \t]+                       { printf(" "); }
.                            { printf("%s", yytext); }

%%

int main(int argc, char **argv) {
    FILE *file;

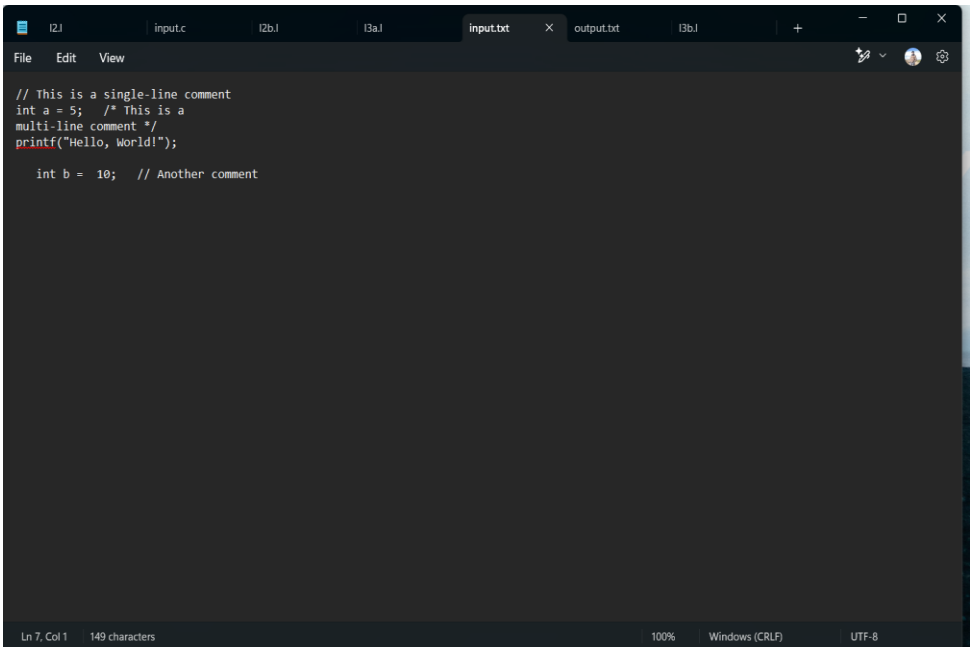
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <inputfile>\n", argv[0]);
        return 1;
    }

    file = fopen(argv[1], "r");
    if (!file) {
        perror("Failed to open file");
        return 1;
    }

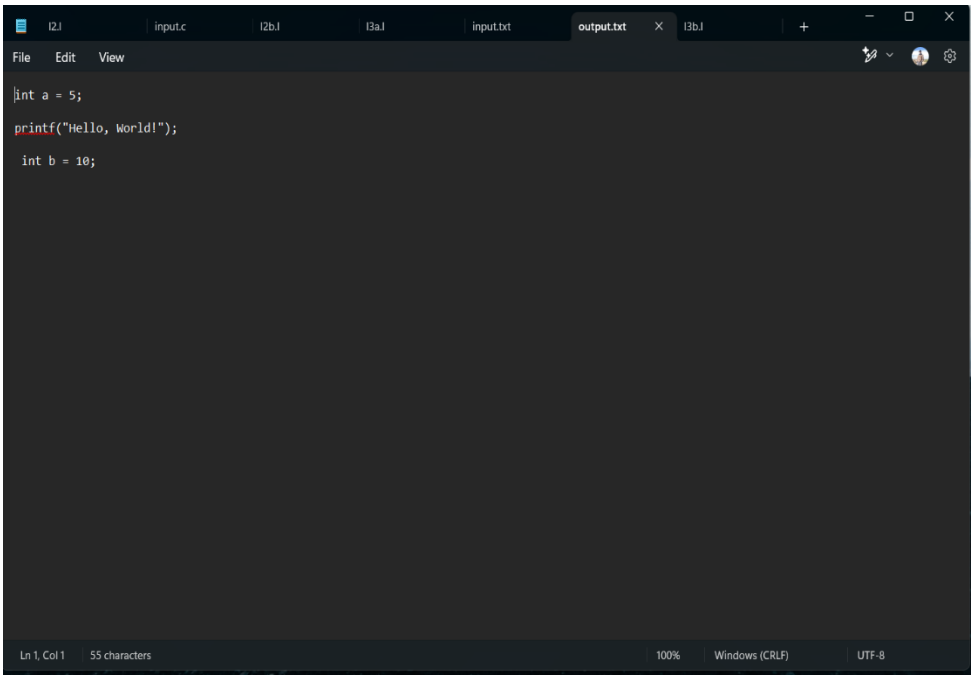
    yyin = file;
    yylex();
    fclose(file);

    return 0;
}
```

Input text code:


l2.l
input.c
l2b.l
l3a.l
input.txt
output.txt
l3b.l
File
Edit
View
// This is a single-line comment
int a = 5; /* This is a
multi-line comment */
printf("Hello, World!");
int b = 10; // Another comment
Ln 7, Col 1
149 characters
100%
Windows (CRLF)
UTF-8

Output:


l2.l
input.c
l2b.l
l3a.l
input.txt
output.txt
l3b.l
File
Edit
View
int a = 5;
printf("Hello, World!");
int b = 10;
Ln 1, Col 1
55 characters
100%
Windows (CRLF)
UTF-8